

ACANTHOSIS NIGRICANS

AT A GLANCE

Acanthosis nigricans is a cutaneous marker, most commonly of insulin resistance and less frequently of genetic disorders and malignancy.

- Characterized by symmetric hyperpigmented, hyperkeratotic, verrucous plaques that bestow a velvety texture on intertriginous skin and occasionally mucocutaneous areas.
- Darker skin pigmentation, insulin resistance, and obesity are more commonly associated with benign acanthosis nigricans.
- Malignant acanthosis nigricans often appears rapidly in older individuals and can involve atypical sites such as mucosal surfaces.

ETIOLOGY AND PATHOGENESIS

Although the precise etiology of benign acanthosis nigricans remains unclear, there is evidence that insulin plays a significant role. The pathogenesis of malignancy-associated acanthosis nigricans is even less clear. A humoral factor produced by the tumor is likely, given cases in which the skin disorder improved or resolved following treatment of the malignancy. Elevated urinary transforming growth factor- α (TGF- α) and increased expression of epidermal growth factor (EGF) receptors in lesional skin were noted in a patient with acanthosis nigricans, acrochordons, Leser-Trélat sign, and melanoma. The enhanced expression of this cytokine and its receptor normalized after removal of the melanoma, and the acanthosis nigricans, acrochordons, and seborrheic keratoses improved postoperatively. A separate case, with gastric cancer, was also shown to express TGF- α and EGF receptors.

More recently, a pathogenic role has been proposed for a tyrosine kinase receptor expressed on the basal cells of the epidermis. In several inherited human skeletal dysplasias associated with acanthosis nigricans, such as craniosynostoses and dwarfing chondrodysplasias, activating mutations in the

fibroblast growth factor receptor 3 (FGFR3) can be observed, whereby FGFR3 has an inhibitory effect on bone formation. In other studies, FGFR3 has oncogenic effects in such malignancies as multiple myeloma, bladder, and cervical carcinoma. With respect to the role of FGFR3 on the epidermis, animal models with activating mutations develop benign epidermal tumors resembling acanthosis nigricans and seborrheic keratoses.

CLINICAL FINDINGS

The clinical hallmark of acanthosis nigricans is development of gray-brown, velvety plaques that may start as a dirty appearance. The hyperpigmentation is later accompanied by hypertrophy, increased skin markings, and papillomatosis. Pruritus can be a problem in some patients. The most commonly involved locations are the axillae, neck, external genitalia, groin, face, inner thighs, antecubital and popliteal fossae, umbilicus, and perianal area. Acrochordons may develop, superimposed on the acanthosis nigricans or on other locations.

Dermatoses with high associations with internal malignancy include tripe palms, Bazex syndrome, paraneoplastic pemphigus, erythema gyratum repens, necrolytic migratory erythema, and hypertrichosis lanuginosa acquisita.

Weakly associated with internal malignancy are acanthosis nigricans, acquired ichthyosis, and multiple seborrheic keratoses.

Management of the dermatosis is the treatment of the malignancy.

Although reported in all age groups, at least 80 percent of malignancy-related acanthosis nigricans occurs in individuals over the age of 40 years.